Hedge Fund Panel Forecasting

Multi-Horizon Time-Series Modelling under a Weighted Skill Score Metric

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Abstract

Abstract to be written after final results have landed. This project studies a panel of approximately 37,000 anonymised time series taken from a Kaggle hedge-fund forecasting competition. The target is forecast at four lookahead horizons under a weighted skill score that concentrates roughly two thirds of the evaluation mass in the top one percent of rows. The first phase of the work, presented in this draft, establishes the structure of the panel, the diagnostic conditions for classical and machine-learning models, and three findings that were not surfaced in the midway report: the four horizons are coupled cumulations of a single short-step process, the maximum panel weight lives at horizon three, and per-horizon target scaling differs by a factor of more than four.

1 Problem Statement

We work with a publicly released Kaggle competition dataset titled *Hedge fund – time-series forecasting.* The competition sponsor releases a panel of anonymised numerical series with engineered features and asks competitors to forecast a continuous target at four lookahead horizons. The dataset is intentionally anonymised: identifiers, feature names, and the underlying financial domain are stripped. What remains is a clean, mathematically well-defined supervised forecasting problem with a non-trivial weighting scheme.

1.1 What is being forecast

Each row of the panel is identified by the tuple

$$(\texttt{code},\ \texttt{sub_code},\ \texttt{sub_category},\ \texttt{horizon},\ \texttt{ts_index})$$

and carries a continuous target y_{target}, a non-negative competition weight, and 86 anonymised numerical covariates of the form $\texttt{feature_a}, \texttt{feature_b}, \ldots$ The integer time index `ts_index` runs over the contiguous range 1–3601 in the public training partition and 3602–4376 in the held-out test partition. The training panel contains 5,337,414 rows and the test panel contains 1,447,107 rows.

The four forecast horizons are $H \in \{1,\ 3,\ 10,\ 25\}$, interpreted as the lookahead in time units. For each $(\texttt{code}, \texttt{sub_code}, \texttt{sub_category})$ the panel typically carries all four horizons at the same `ts_index`, so the natural unit of analysis is the 4-tuple key above. There are 36,923 distinct series in train under that definition.

1.2 Why the problem is hard

The dataset combines three well-known difficulties that make naive baselines a bad guide:

1. **Heavy-tailed target.** The target distribution is centred near zero with an interquartile range of approximately ± 0.05, but the full range runs from roughly -2200 to $+2300$. The 1% and 99% percentiles are at approximately ± 82. Plain mean-squared losses are therefore extremely sensitive to a small number of tail rows.

2. **Concentrated evaluation weight.** The competition metric is sample-weighted, and the weights are extraordinarily skewed: the median weight is approximately 1.7×10^3 while the maximum is 1.4×10^{13}. As a result, the top one percent of rows by weight carry approximately 64% of all evaluation mass, and the top five percent carry approximately 92%. Any unweighted analysis can recommend the wrong model.

3. **Limited per-series history at long horizons.** Although the panel is large in aggregate, only about 5% of individual series cleanly cross the validation cutoff used in this work (`ts_index` = 2880). Some series have as few as seven training points before the cutoff, which makes per-series classical fitting infeasible for many model families.

1.3 Why the problem is interesting

Beyond being a real competitive forecasting problem, three properties make the dataset a useful subject for a methodological course project:

- *Anonymisation forces statistical reasoning.* Without semantic feature names we cannot fall back on domain priors. Every claim must be supported by a measurable property of the data.

- *The weighting scheme rewards precision on rare rows.* A model that improves average behaviour while degrading on the high-weight rows can lose to a simpler model. This is a mechanically different objective from standard regression and warrants explicit study.

- *The four horizons are coupled.* As we show in Section 4.16, the targets at $H \in \{3, 10, 25\}$ are well approximated by rolling cumulative sums of the $H = 1$ process. This structure is not advertised in the dataset description and admits a modelling strategy that the midway report did not consider.

1.4 Scope of this report

The remainder of this report covers the diagnostic and exploratory phase of the project. Section 2 formalises the panel keys and the chronological constraint that any validation strategy must respect. Section 3 develops the official competition metric in detail, with worked numerical examples and a comparison against standard error metrics. Section 4 carries out a full exploratory analysis: schema sanity, missingness, target and weight distributions, stationarity diagnostics, autocorrelation structure, feature audit, and three findings about the multi-horizon structure of the panel that constrain every modelling decision in subsequent chapters of the project. Modelling chapters (classical baselines, global gradient-boosted models, the multi-horizon aggregation experiment, probabilistic forecasting, and ensembles) will be added as those notebooks land.

2 Forecast Targets and Series Identity

A panel forecasting problem only makes sense once the unit of analysis is unambiguous. This section fixes the definitions used throughout the project.

2.1 Panel keys

A row of the panel has primary keys (`code`, `sub_code`, `sub_category`, `horizon`, `ts_index`). We treat the 4-tuple

$$s \equiv (\texttt{code}, \texttt{sub_code}, \texttt{sub_category}, \texttt{horizon})$$

as the natural *series identifier*, and treat `ts_index` as the time axis. There are 23 codes, 180 distinct sub-codes, 5 sub-categories, and 4 horizons in train, yielding 36,923 distinct series. Test uses a narrower sub-code support of 47 distinct values, which is a coverage shift rather than a schema mismatch (every test category level already exists in train).

2.2 The four horizons are coupled

A naive reading of the schema would treat the four horizons of the same (`code`, `sub_code`, `sub_category`) as four independent series, simply distinguished by an extra integer label. Two empirical facts in Section 4 contradict this view:

1. For approximately 1.21 million distinct (`code`, `sub_code`, `sub_category`, `ts_index`) tuples, all four horizons are simultaneously present (Section 4.15).

2. For the same triple, the target at horizon k is closely approximated by the rolling sum of the $H = 1$ target over the next k steps:

$$y_t^{H_k} \approx \sum_{i=0}^{k-1} y_{t+i}^{H_1}.$$

   We verify this on a stratified sample of 500 triples in Section 4.16, and find median Pearson correlations of 0.96, 0.94 and 0.91 at $k = 3, 10, 25$ respectively.

The practical consequence is that fitting four independent regressors per triple discards a strong constraint that the data construction itself imposes. We exploit this constraint in the H1-aggregation experiment scheduled for a later notebook.

2.3 Chronological constraint

The training partition runs over `ts_index` $\in [1, 3601]$ and the test partition over $[3602, 4376]$. Because test is a strict temporal suffix of train, any internal validation strategy must also be chronological, with all training points strictly preceding the validation points in time. Random or leave-one-series-out cross-validation would mix future information into earlier states and overstate generalisation.

We use a single canonical chronological partition throughout the project, fixed in the shared utilities at `cf.splits`:

- **Train:** `ts_index` ≤ 2880.
- **Meta** (used only for fitting ensemble weights): $2880 <$ `ts_index` ≤ 3240.
- **Holdout** (final scored window): $3240 <$ `ts_index` ≤ 3601.
- **Test** (`ts_index` ≥ 3602): unlabeled, used only for submission-style sanity checks.

The cutoff `ts_index` $= 2880$ matches the cutoff used in the midway report, preserving comparability with prior diagnostic figures. The meta/holdout boundary at 3240 places approximately one third of the post-cutoff window into the meta slice, leaving the remaining two thirds for final scoring.

2.4 Eligibility for per-series studies

Some classical model families (ARIMA, ARMA, MA, AR with order ≥ 2) require a minimum training history. For per-series diagnostic studies we restrict attention to series that both cross the train/validation cutoff and have at least 120 training rows. There are 1,930 such *eligible* series in the panel, roughly 5% of the 36,923 total. Global pooled models and the H1-aggregation experiment use every row whose `ts_index` falls inside the relevant split, so they are not constrained by per-series length.

3 The Weighted Skill Score

The competition is decided by a single scalar: the *weighted skill score*. Because the rest of the project uses this metric for every model comparison, ensemble weight, and bootstrap interval, it deserves a dedicated section. We give the formula, walk through every term, work three numerical examples, and explain why a competition organiser would pick a metric of this shape over more familiar alternatives such as RMSE, MAE, or R^2.

3.1 Definition

Given truths y_i, predictions $\hat{y}_i$, and non-negative weights w_i for $i = 1, \ldots, n$, the weighted skill score is

$$\text{score}(y, \hat{y}; w) \;=\; \sqrt{1 \;-\; \text{clip}_{[0,1]}\left(\frac{\sum_i w_i\,(y_i - \hat{y}_i)^2}{\sum_i w_i\, y_i^2}\right)} \tag{1}$$

where $\text{clip}_{[0,1]}(x) = \min(\max(x, 0), 1)$. The function returns 1 for a perfect prediction and 0 when the prediction is no better than the all-zero forecast.

3.2 Term-by-term walkthrough

Numerator $\sum_i w_i(y_i - \hat{y}_i)^2$. This is the weighted sum of squared residuals. It is zero when $\hat{y} = y$ everywhere and grows quadratically with prediction error. Because the squared error is multiplied by w_i, rows with larger weights contribute proportionally more to the penalty. In our panel the weight distribution is extremely concentrated, so this term is dominated by a small fraction of high-weight rows.

Denominator $\sum_i w_i y_i^2$. This is the weighted "energy" of the truth. It is purely a function of the data, independent of the prediction, and acts as a normaliser. Without this normaliser, two panels with very different target magnitudes (for example a high-volatility horizon and a low-volatility horizon) could not be scored on the same scale.

The ratio. The ratio

$$R \;=\; \frac{\sum w_i(y_i - \hat{y}_i)^2}{\sum w_i y_i^2}$$

is a weighted, non-centred "unexplained variance" measure. It is 0 when predictions are perfect and 1 when predictions are identically zero. It can exceed 1 when predictions are worse than zero (for example, when the prediction has the wrong sign at scale). It can in principle be unbounded above if the prediction is far from any reasonable point estimate.

The clip. $\text{clip}_{[0,1]}(R)$ enforces $R \in [0, 1]$. The lower bound at zero is automatic from the squares; the upper bound at 1 is the meaningful one. It says: *no model that is worse than predicting zero gets credit for being even worse.* The clip stops the inside of the square root from going negative and ensures the score is always real. Operationally the clip rarely engages on competent models, but it makes the metric robust to a single catastrophic prediction.

The square root. The outer square root maps $1 - R$ from a variance-like quantity onto a correlation-like scale. A skill of 0.5 corresponds to $R = 0.75$, that is, "three quarters of weighted target energy is unexplained". A skill of 0.95 corresponds to $R = 0.0975$, "ten percent of energy unexplained". The square root makes mid-range skills feel as familiar as Pearson correlations rather than as compressed variance ratios.

3.3 Worked numerical examples

To make the behaviour of Equation (1) concrete, consider three predictions on a tiny five-row panel where the truth and weights are

$$y = (1, 2, 3, 4, 5)^\top, \qquad w = (1, 1, 1, 1, 10)^\top.$$

Note the heavy concentration: the last row carries 10/14 of the total weight. The weighted target energy is $\sum w_i y_i^2 = 1 + 4 + 9 + 16 + 250 = 280$.

Table 1: Behaviour of the weighted skill score on three contrasting predictions.

Prediction $\hat{y}$	Numerator	Ratio R	clip(R)	Skill
$(1, 2, 3, 4, 5)$ — perfect	0	0.000	0.000	1.000
$(0, 0, 0, 0, 0)$ — zero	280	1.000	1.000	0.000
$(0.9, 1.9, 3.1, 4.0, 4.9)$ — close	0.13	0.000	0.000	0.999
$(2, 4, 6, 8, -5)$ — wrong sign tail	1014	3.621	1.000	0.000

A few things are worth noticing in Table 1. First, predicting all zeros lands exactly at skill = 0, the natural reference point. Second, a prediction that is wildly wrong on the high-weight last row is clipped to skill = 0; the metric does not pay attention to how much worse than zero the model is. Third, a small-error model on the four low-weight rows but a slightly larger error on the high-weight row could already drop the skill noticeably, because the high-weight row dominates the numerator.

3.4 Comparison against standard regression metrics

Table 2 contrasts the weighted skill score with the metrics most often reported in regression problems on the same five-row toy example.

Table 2: Behaviour of standard metrics versus the weighted skill score on the same toy panel.

Prediction $\hat{y}$	RMSE	weighted RMSE	R^2	Skill
$(1, 2, 3, 4, 5)$	0.000	0.000	1.000	1.000
$(0, 0, 0, 0, 0)$	3.317	4.472	−0.367	0.000
$(0.9, 1.9, 3.1, 4.0, 4.9)$	0.114	0.097	0.999	0.999
$(2, 4, 6, 8, -5)$	5.200	8.510	−3.36	0.000

The skill score and weighted RMSE rank the four predictions in the same order, but they differ in two important ways. First, skill is bounded in $[0, 1]$, which is easier to read on a competition leaderboard. Second, R^2 uses the centred denominator $\sum w_i (y_i - \bar{y})^2$ while skill uses the non-centred denominator $\sum w_i y_i^2$; this makes skill more meaningful when the target is genuinely centred near zero (as it is in our panel) and avoids the embarrassing situation where R^2 goes negative for predictions worse than the constant mean.

3.5 Why a competition organiser would design this metric

Five properties of Equation (1) explain why a metric of this shape is sensible for a public-leaderboard hedge-fund forecasting task:

1. *Reference-relative.* Different series in a hedge-fund panel can have wildly different volatilities. Plain RMSE will always rank a model better on a low-volatility series than on a high-volatility series, even when the relative quality is identical. Dividing the squared error by the weighted target energy neutralises this scale effect, so a 1% relative error on a 0.0001-scale series and a 1% relative error on a 100-scale series contribute proportionally.

2. *Beat-the-zero baseline.* Predicting $\hat{y} = 0$ identically gives skill $= 0$. Any positive skill therefore signals "the model has learned something". This threshold is a more honest baseline than "beat the naive lag-1 forecast", which can be hard to define cleanly across heterogeneous series.

3. *Bounded* $[0, 1]$ *via the clip and square root.* A bounded metric is leaderboard-friendly: scores compress into a familiar 0–1 range, making rankings easy to display and small differences easy to detect. The clip in particular makes the metric robust to an occasional catastrophic prediction, so competitors are not penalised disproportionately for one outlier row.

4. *Sample-weighted.* The weights let the organiser encode importance. In our panel, the weights are concentrated on a small number of rows, plausibly reflecting financial significance such as assets under management or notional traded size. A model that performs well on rows that "matter financially" is valued more than one that performs equally well across all rows.

5. *Interpretable on the correlation scale.* The square root puts the score on roughly the same scale as a Pearson correlation, which most readers find intuitive. Without the square root, mid-range skills would compress; with it, the values 0.3, 0.5, 0.7, 0.9 carry the rough qualitative meaning that practitioners assign to correlations.

3.6 Practical consequences for this project

Equation (1) is implemented once in the shared utility `cf.metrics.weighted_skill_score` and used by every later notebook for both model selection and final scoring. The function `cf.metrics.bootstrap_skill` returns 95% confidence intervals by row resampling; we report bootstrap intervals alongside every leaderboard number to avoid declaring winners that are within sampling noise of each other.

Two structural observations carry through to every subsequent modelling decision:

- **High-weight rows decide the leaderboard.** Because approximately 64% of weight sits in the top 1% of rows (Section 4.5), the score is essentially a measure of how well the model predicts those few rows. We must therefore train with sample weights, not just evaluate with them.

- **Plain RMSE will lie.** A model that improves average behaviour at the cost of a small degradation on a high-weight row can lose to a simpler model. We never report unweighted RMSE as a primary headline number.

4 Dataset and Exploratory Analysis

This section reports the exploratory analysis of the panel before any modelling. The aim is twofold. First, to establish the basic statistical shape of the data so that later modelling choices

can be motivated rather than guessed. Second, to surface three structural findings about the panel — multi-horizon coupling, the cumulation hypothesis, and per-horizon weight dominance — that the midway report did not document but that materially change the modelling space. All numerical results below are produced by notebook `00_data_structure.ipynb` and the corresponding figures are saved as both PNG and PDF assets.

4.1 Schema and panel structure

The training panel contains 5,337,414 rows, organised by 23 distinct `code` values, 180 distinct `sub_code` values, and 5 `sub_category` values, across 4 horizons. The integer time index `ts_index` runs over the contiguous range $[1, 3601]$. The test partition continues from `ts_index` $= 3602$ to 4376 with 1,447,107 rows. Treating the 4-tuple (`code`, `sub_code`, `sub_category`, `horizon`) as the natural series identifier yields 36,923 distinct series.

4.2 Train versus test comparability

The only columns present in train but not in test are y_{target} and `weight`, which is exactly what we expect (test carries no labels or evaluation weights). Every category level seen in test exists in train, so there is no cold-start schema issue. The sub-code support narrows from 180 distinct values in train to 47 in test, which is a coverage shift rather than a schema mismatch. We must respect this when interpreting per-sub-code conclusions, but it does not require any change to the core modelling setup.

4.3 Missing values

Missingness is computed on a 500,000-row stratified sample to keep memory in check. Of the 86 anonymised features, 48 contain at least one missing value, but the maximum missing rate across all columns is approximately 12.5%. Figure 1 shows the top fifteen offenders.

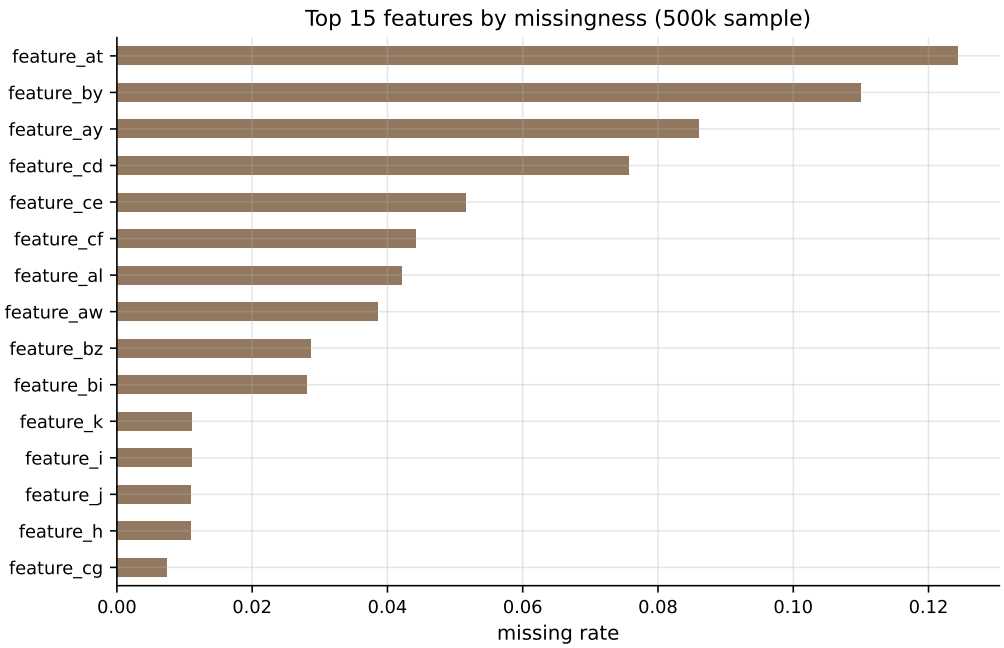

Figure 1: Top fifteen features by missingness, on a 500,000-row stratified sample of train. The maximum missing rate is approximately 12.5%, low enough that median imputation inside a fold is sufficient and there is no need to drop columns or use complex imputers.

Reading. The missing-rate magnitudes are non-trivial but bounded. A simple within-fold median imputation handles every column without leakage. We therefore do not develop a separate imputation strategy.

4.4 Target distribution

The target y_{target} is centred very close to zero (median -0.0006, IQR ± 0.05), but has extremely heavy tails. The 1% and 99% quantiles sit at approximately -82.8 and $+62.9$ respectively, while the full range runs from approximately -2202 to $+2314$.

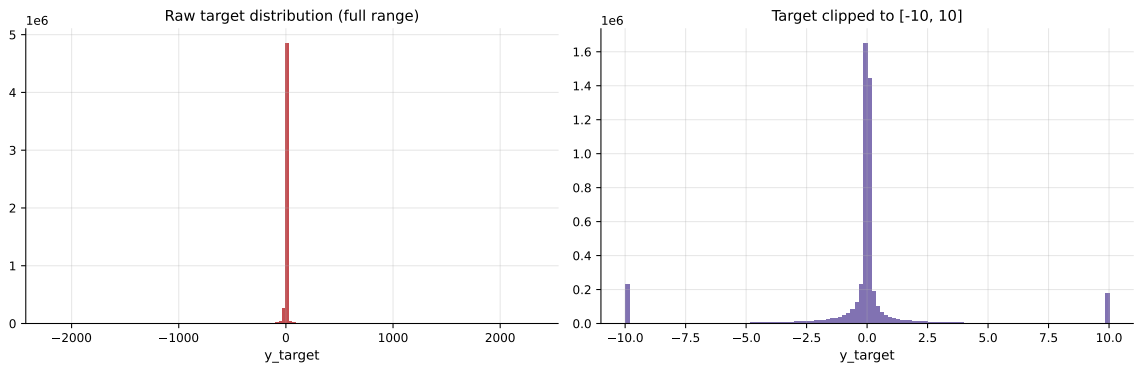

Figure 2: Raw target distribution (left) and the same distribution clipped to $[-10, 10]$ (right). Most of the mass concentrates near zero, with rare extreme tails up to the order of ± 2200.

Reading. A target with this shape disqualifies plain unweighted MSE as a sensible loss for any model that has to be robust across the full panel. The tails are real outcomes, not data corruption, so we cannot just clip them. The combination of heavy tails and the weighting scheme (next subsection) is the central source of methodological care in this project.

4.5 Weight distribution and metric concentration

The competition weights are extraordinarily skewed. The median weight is 1,699, while the maximum is 1.39×10^{13}. Figure 3 shows the weight distribution clipped at the 99th percentile (left) and the cumulative-share Lorenz-style curve (right). The top 1% of rows carry 64.2% of total weight; the top 5% carry 92.6%; the top 10% carry 98.3%.

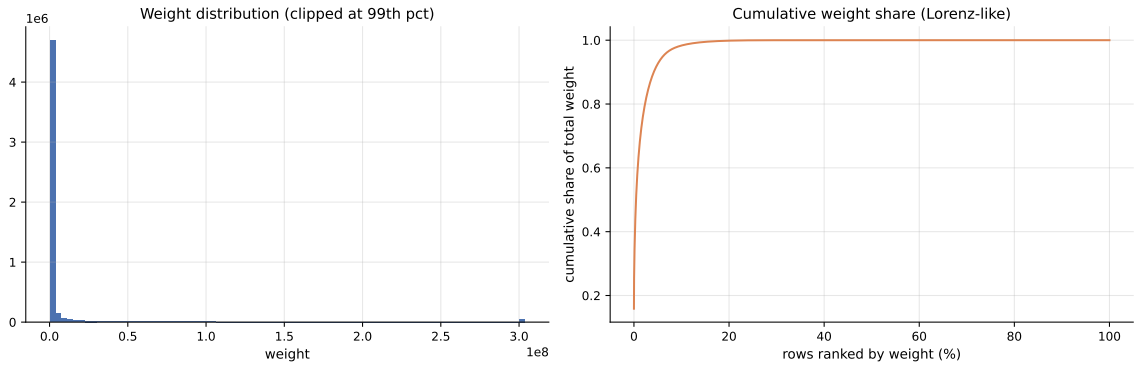

Figure 3: Weight distribution clipped at the 99th percentile (left) and cumulative share of total weight ranked by weight (right). Approximately 64% of total weight is concentrated in the top 1% of rows; 92% in the top 5%.

Reading. Any evaluation that ignores weights describes a different competition. Plain RMSE, unweighted MAE, and naive cross-validation can recommend the wrong model family. The weighted skill score formalised in Section 3 is the only metric we report as a headline. Operationally, a model that accurately predicts the top one percent of rows has more leverage on the leaderboard than a model that is uniformly mediocre across all rows.

4.6 Series coverage and split feasibility

Not every series has enough history to support time-based validation. Figure 4 shows the distribution of per-series lengths. Out of 36,923 distinct series, only 1,930 (approximately 5%) cleanly cross the train/validation cutoff at `ts_index` = 2880. Many series end before the cutoff, contributing zero validation rows.

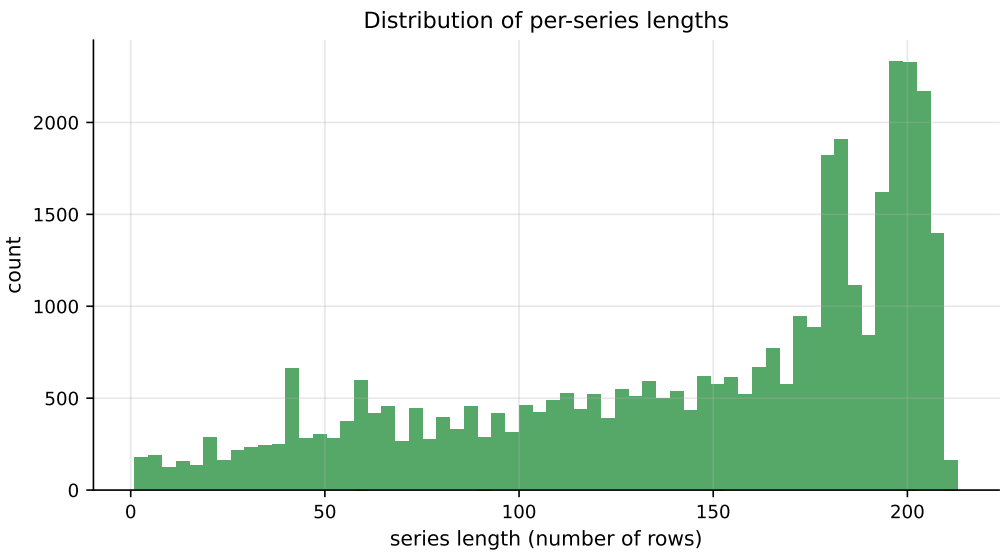

Figure 4: Distribution of per-series lengths. Most series are short, and the tail of long series carries the structural signal that classical fits can exploit.

Reading. Per-series classical fitting is feasible only on the eligible subset (cross the cutoff and have at least 120 training rows). Global pooled models are not constrained by per-series length and can use every row whose `ts_index` falls inside the relevant split. This asymmetry is one of the reasons the comparison between local classical models and global ML models is not perfectly apples-to-apples.

4.7 Representative-series selection

For every per-series figure in this project we plot the same four series. They are selected with explicit reasons rather than at random, in order to span four distinct stress regimes: long history, large weight, high variability, and near-flat behaviour. Table 3 lists the four chosen series. They match the selection used in the midway report, preserving visual continuity.

Figure 5 shows the raw y_{target} trajectory of each chosen series with the train/validation cutoff marked. The four panels look profoundly different: the most-volatile series swings between -500 and $+500$, while the most-stable series occupies a numerical range narrower than 0.001.

Reading. The four panels make it visually clear that a one-size-fits-all model will struggle. The most-stable series is essentially numerical noise; the most-volatile series has obvious cyclical

Table 3: Four representative series chosen for per-series studies. Each series probes a different regime of the panel.

Reason	Series key (code/sub_code)	Horizon	Length
Longest history	X9BZ68VQ / OYJGNSQK / DPPUO5X2	1	212
Highest total weight	SJZP0OVU / OYJGNSQK / NQ58FVQM	25	156
Most volatile	W4S29LF4 / KL66VIS3 / PHHHVYZI	25	162
Most stable	SJZP0OVU / OYJGNSQK / NQ58FVQM	1	170

structure. The pattern of forecast performance across these four panels is one of the cleanest diagnostics for whether a given model family is generalising or just fitting one regime well.

4.8 Stationarity diagnostics

Classical ARIMA-family forecasting assumes stationarity, or stationarity-after-differencing. We therefore run the Augmented Dickey–Fuller test on a stratified sample of 200 eligible series. The ADF test takes the null hypothesis that the series has a unit root (non-stationary); we reject at the 5% level when the p-value is below 0.05. Figure 6 summarises the results.

Approximately 70.5% of sampled series reject the unit-root null at the 5% level. The picture is more nuanced when broken down by horizon: H1 series are more often stationary than longer-horizon cumulative series, which is consistent with the multi-horizon coupling we will document in Section 4.16.

ADF and KPSS have opposite null hypotheses, so running both gives a more honest joint verdict: ADF takes non-stationarity as the null, while KPSS takes stationarity as the null. We classify a series as *stationary* when ADF rejects and KPSS does not, as *unit root* when ADF fails to reject and KPSS rejects, and as *inconclusive* otherwise. Figure 7 reports the joint distribution.

The stationary share dominates, but the inconclusive bucket is large enough that one-size-fits-all claims about raw levels would be unsafe. We need a repair step (differencing) for the unit-root and inconclusive cases.

Figure 8 re-runs the ADF test on the first-differenced series. The stationary share rises from 70.5% on raw levels to 98.5% after differencing.

Reading. The diagnostic chain motivates ARIMA with $d = 1$ as a sensible default for the classical study, and it explains why differenced features are worth engineering for the global ML models. The fact that 98.5% of series become stationary after a single difference also bears on the cumulation finding in Section 4.16: differencing the higher-horizon target partially inverts the rolling-sum construction back into the H1 process.

4.9 Visual stationarity — rolling statistics

Statistical tests can disagree on short or noisy series. A rolling plot is the simplest visual check: a flat rolling mean indicates constant-mean behaviour; a flat rolling standard deviation indicates constant variance. Figure 9 shows the four chosen series, raw, with a 24-step rolling mean and rolling std overlaid.

Figure 10 repeats the same diagnostic on the first-differenced versions of the same series. The rolling means hug zero and the rolling standard deviations flatten, visually confirming the statistical jump in the previous subsection.

Reading. The visual evidence agrees with the formal tests: differenced series behave more like white noise, raw series often drift. This is exactly the visual behaviour we want before

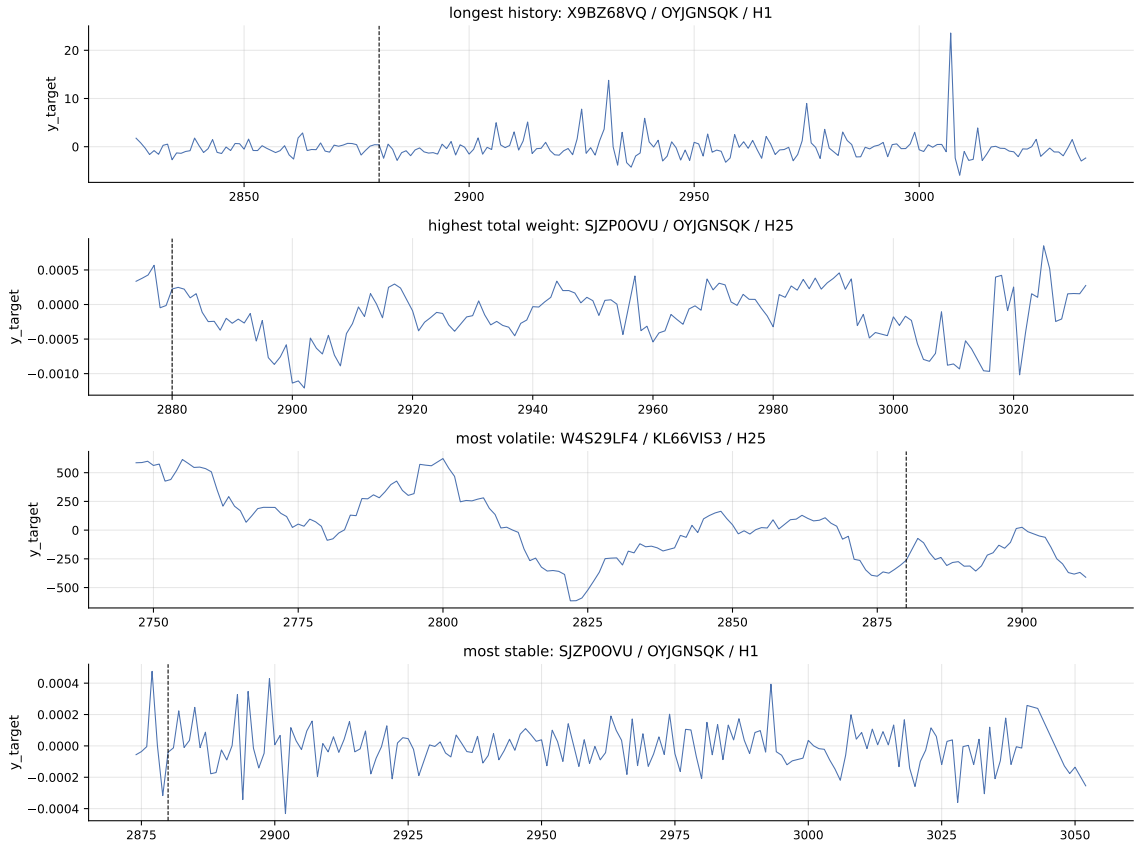

Figure 5: Raw y_{target} for the four chosen representative series. The dashed vertical line marks `ts_index` = 2880, the train/validation cutoff. The four panels span the four stress regimes that any model has to handle.

fitting a differenced classical model.

4.10 Lag structure — ACF and PACF

The autocorrelation function (ACF) shows correlation between the series and lagged copies of itself; the partial autocorrelation function (PACF) isolates each lag's direct contribution after controlling for shorter lags. A sharp PACF cutoff suggests an autoregressive order; a sharp ACF cutoff suggests a moving-average order. Figure 11 shows the ACF and PACF of the longest-history anchor series.

A single series can mislead. We aggregate the absolute ACF and PACF across all 200 sampled series in Figure 12. The panel-level view reveals which lags are reliably informative across the panel rather than artefacts of a single trajectory.

Reading. The panel ACF/PACF tell us two things. First, short-memory dynamics are the dominant signal: lags 1–3 carry the bulk of the structure, with rapid decay thereafter. Second, the absence of seasonal bumps argues against building the modelling around seasonality. The lag feature set we engineer for the global ML models therefore prioritises lags 1, 2, 3, 5, 10 and skips seasonal indicators.

4.11 STL decomposition

Seasonal-Trend decomposition using LOESS (STL) splits a series into trend, seasonal, and residual components. We apply STL to the longest-history anchor with a period of 24 and observe the relative amplitudes of the three components. Figure 13 shows the result.

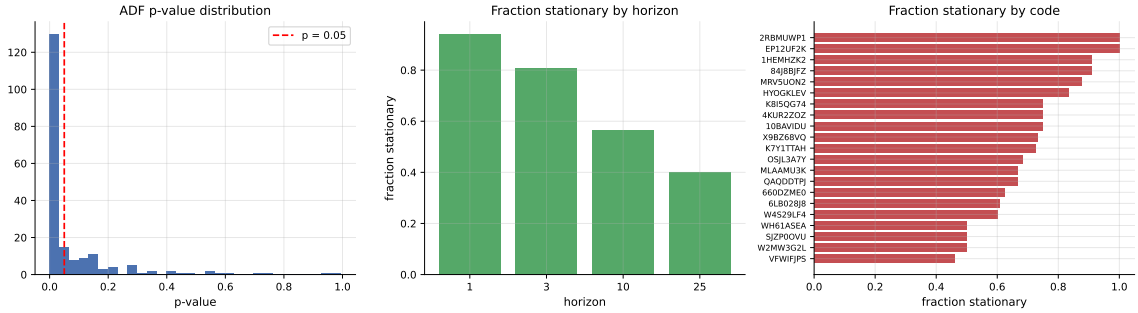

Figure 6: ADF results on 200 sampled eligible series. Left: p-value distribution with the 0.05 threshold marked. Centre: fraction stationary by horizon. Right: fraction stationary by code.

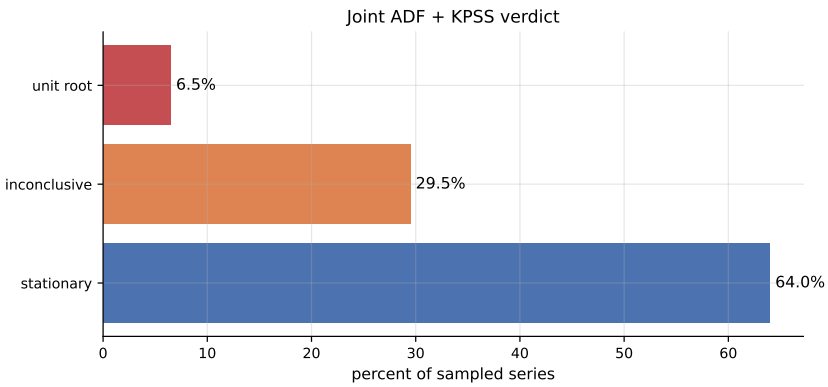

Figure 7: Joint ADF + KPSS verdict on 200 sampled series. Stationary 64.0%, inconclusive 29.5%, unit root 6.5%.

Reading. The trend component captures the slow level drift that ADF picked up, but the seasonal amplitude is small relative to residual standard deviation. Combined with the absence of seasonal bumps in the panel ACF, this confirms that strong periodic structure is not a property of this panel. Modelling effort should go into short-memory dynamics rather than seasonal indicators.

4.12 Feature audit

The 86 `feature_*` columns are anonymised and we cannot interpret them by name. We can, however, audit them statistically. On a 200,000-row stratified sample we compute Pearson correlation between each feature and y_{target}, then rank features by absolute correlation. Figure 14 shows the top twenty.

Reading. The single largest absolute correlation is below 0.10. No single feature predicts the target well on its own. This rules out a univariate model and warns against placing too much weight on any one feature in importance plots, but it does not mean the features are useless: gradient-boosted trees and other non-linear ensembles can combine many weak features into a meaningful signal.

4.13 Feature multicollinearity

If the top features are highly correlated with each other, their importance ranking is ambiguous — they may share signal rather than each contributing independently. Figure 15 shows a correlation heatmap among the top twenty features and the target.

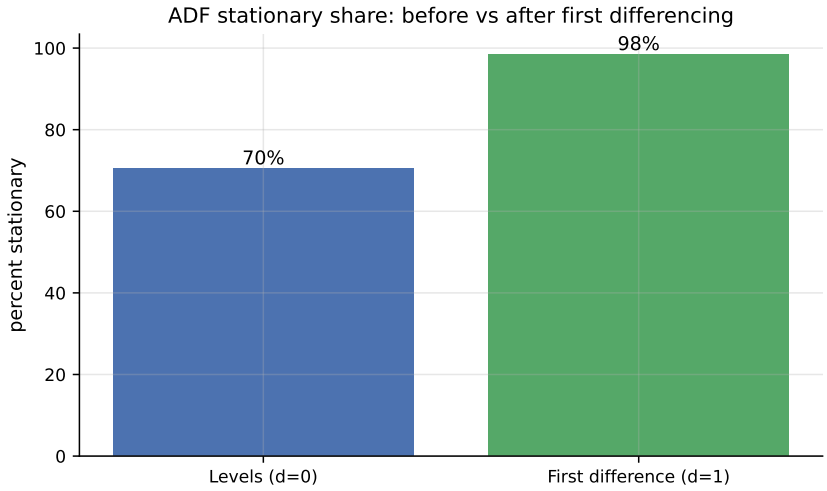

Figure 8: ADF stationary share at the 5% level: levels (left, $d = 0$) versus first differences (right, $d = 1$). Differencing pushes the share to nearly 100%.

Reading. The heatmap shows three things. The y_{target} row and column confirm the modest marginal signal seen in the previous subsection. Strong feature–feature correlation blocks reveal real redundancy: multiple features carry overlapping information. Naive feature-importance rankings should therefore be read with collinearity in mind. We will not pretend that a feature ranked first in importance is uniquely responsible for the model's behaviour.

4.14 Cross-series relationships

Within the same `code`, different sub-codes might share signal. If they do, global pooled models can borrow strength across related sub-codes; per-series local models cannot. We compute lagged cross-correlations between random pairs of sub-codes within the same `(code, horizon)` group. Figure 16 shows six representative pairs.

Reading. Several pairs show clear peaks away from lag 0, indicating real lead/lag relationships within a code. This is exactly the structure that global models exploit: training a single model on every sub-code lets the model learn the shared component, while per-series fitting wastes that signal entirely. The cross-series evidence is one of our strongest a priori arguments for choosing a global ML approach over the per-series classical approach pursued in the midway.

4.15 Multi-horizon coupling

The midway report treated (`code`, `sub_code`, `sub_category`, `horizon`) as the natural series identifier and fit each horizon independently. The exploratory analysis in this project shows that the four horizons of the same triple are not independent. For approximately 1.21 million distinct (`code`, `sub_code`, `sub_category`, `ts_index`) tuples, all four horizons are simultaneously present.

A second consequence of multi-horizon coupling is per-horizon target scaling. Figure 17 shows the target standard deviation and maximum row weight by horizon.

The target standard deviation rises from approximately 11.7 at H1 to 52.8 at H25, a factor of 4.5.

Reading. This finding has direct modelling consequences. A single model trained on horizons pooled into one DataFrame, without horizon-aware standardisation, will be dominated by H25 magnitudes and predict at H25 scale. Predictions at H25 scale evaluated against H1 targets will

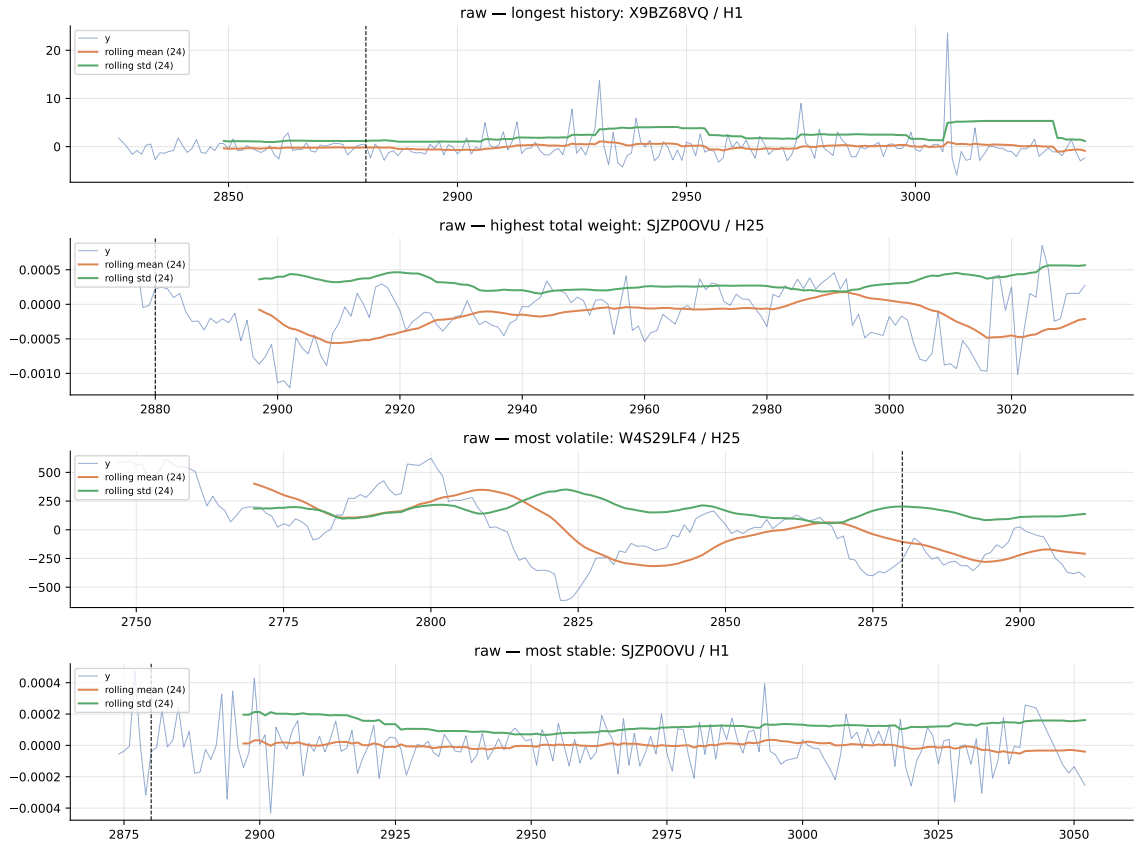

Figure 9: Rolling mean and rolling standard deviation (window = 24) on the raw four representative series. The most-volatile series shows clear drift in level and bulging variance.

overshoot by a factor of roughly 4.5, producing catastrophically large weighted RMSE on H1 rows. This is exactly the failure mode observed in earlier ML artefacts attached to the project — the original XGBoost run reported a weighted RMSE more than two hundred times the naive baseline, consistent with a per-horizon-scale leak. The remedy is straightforward: train one model per horizon, or normalise the target by horizon before training.

4.16 Cumulation hypothesis

Given the multi-horizon coupling above, the natural hypothesis about how the four horizons relate to each other is that the target at horizon k is a cumulative aggregate of the H1 target over the next k steps:

$$y_t^{H_k} \approx \sum_{i=0}^{k-1} y_{t+i}^{H_1}. \tag{2}$$

We test Equation (2) on a stratified sample of 500 triples that have all four horizons and at least 60 H1 rows. For each triple we pivot the data so that horizons sit on columns, compute the rolling sum of the H1 series over the appropriate window, and report the Pearson correlation between $y_t^{H_k}$ and that rolling sum. Figure 18 shows the distribution of correlations and the median ratio.

Table 4 reports the summary statistics. At $k = 3$ the median Pearson correlation is 0.96 and the median ratio is 1.00, indicating essentially exact aggregation. The relation is slightly noisier at $k = 10$ and $k = 25$, with median ratios of 0.95 and 0.91, but still clearly dominant.

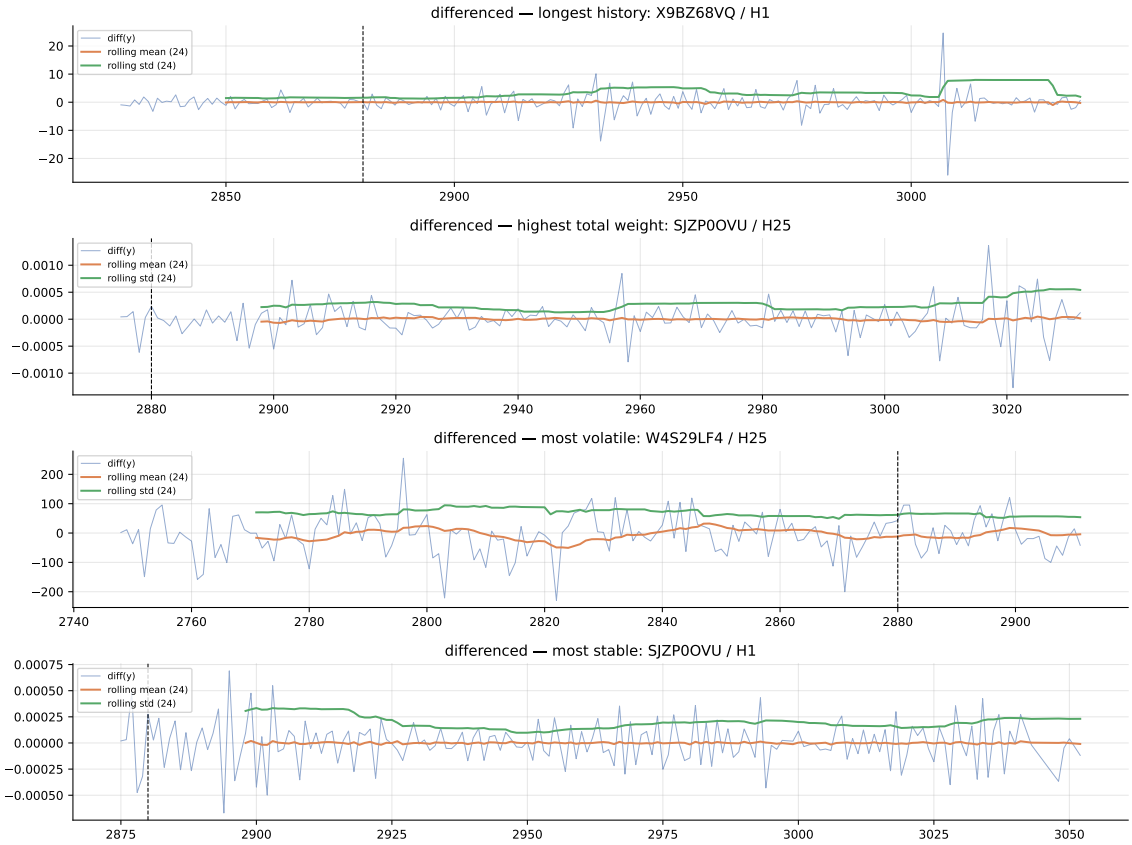

Figure 10: Rolling mean and rolling standard deviation (window = 24) on the first-differenced four representative series. The mean now oscillates closer to zero and the variance becomes more stable.

Table 4: Cumulation hypothesis test summary on 500 triples.

k	n triples	mean corr	median corr	median ratio
3	500	0.92	0.96	1.00
10	500	0.86	0.94	0.95
25	500	0.78	0.91	0.91

Reading. Two consequences for the modelling work that follows. First, fit-H1-and-aggregate becomes a viable, possibly superior strategy to fitting per-horizon models independently: a single H1 model gives H3, H10, and H25 forecasts for free, and the aggregation is mathematically forced by the data construction. We test this in a dedicated H1-aggregation experiment in a later notebook. Second, differencing operations on $y^{H_{25}}$ partially recover the y^{H_1} process, which explains why ARIMA with $d = 1$ wins the most-volatile H25 series in the midway classical study: differencing inverts the cumulation. The midway interpretation that ARIMA was capturing a difficult pattern is partly correct; it was also undoing the cumulative construction.

4.17 Per-horizon weight dominance

The weight distribution analysis in Section 4.5 was panel-wide. Because `weight` and `horizon` are correlated, the rows that drive the weighted skill score may not be evenly distributed across horizons. Figure 19 reports the share of total panel weight per horizon and the share of weight within the top 1% of all weight mass.

The headline numbers are striking. The maximum panel weight (1.39×10^{13}) is at H3, not

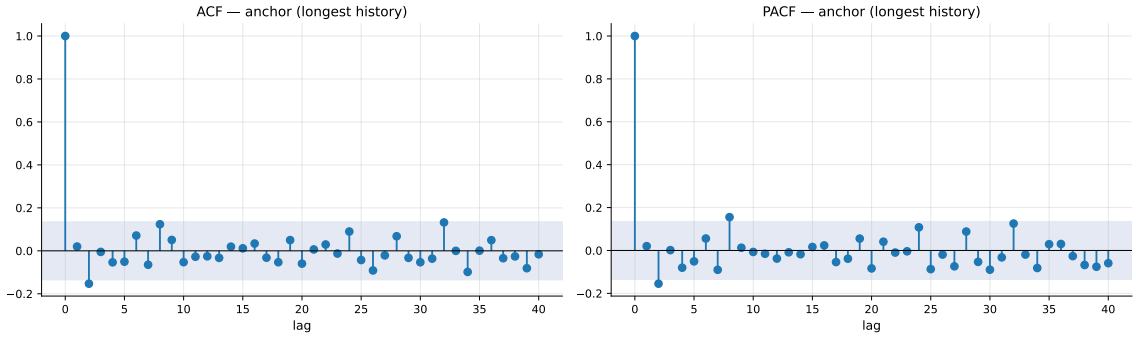

Figure 11: ACF (left) and PACF (right) of the longest-history anchor series, lags 0–40. The shaded band shows the 95% confidence interval under white noise.

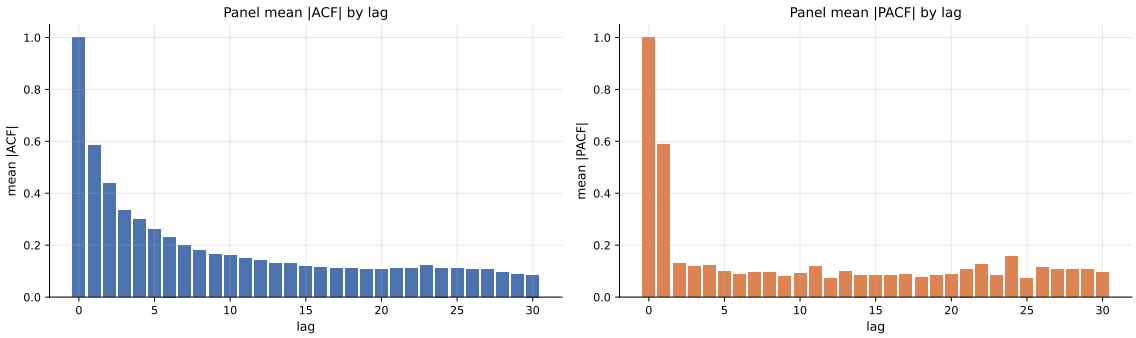

Figure 12: Panel-level mean |ACF| (left) and mean |PACF| (right) across 200 sampled eligible series. Persistent low-order lags dominate; higher lags decay quickly. There are no clean seasonal bumps.

H1. H3 alone holds 37.4% of total panel weight and 43.6% of the top-1% weight mass; H1 holds 25.4%, H10 holds 14.3%, and H25 holds 16.8%.

Reading. A model that predicts H3 well on the small set of high-weight rows will tend to win the leaderboard regardless of how it performs on the other horizons. This was missed in the midway report. The practical consequence is that we report H3 weighted skill as a separate headline number alongside the panel-wide score, and we track H3 performance carefully when comparing models.

4.18 EDA-driven modelling implications

The EDA constrains modelling along several axes. Table 5 summarises the constraints and the corresponding decisions for later notebooks.

These constraints shape every modelling chapter that follows. The classical baselines (notebook 01) honour the chronological cutoff and the four representative series. The global LightGBM models (notebook 02) train per horizon with sample weights. The H1-aggregation experiment (notebook 03) operationalises the cumulation hypothesis. The probabilistic forecasting work (notebook 04) addresses the heavy tails and weak per-feature signal directly. The ensembles (notebook 05) fit weights on a meta slice that respects chronology. The final comparison (notebook 06) reports both panel-wide and H3-specific weighted skill scores with bootstrap intervals.

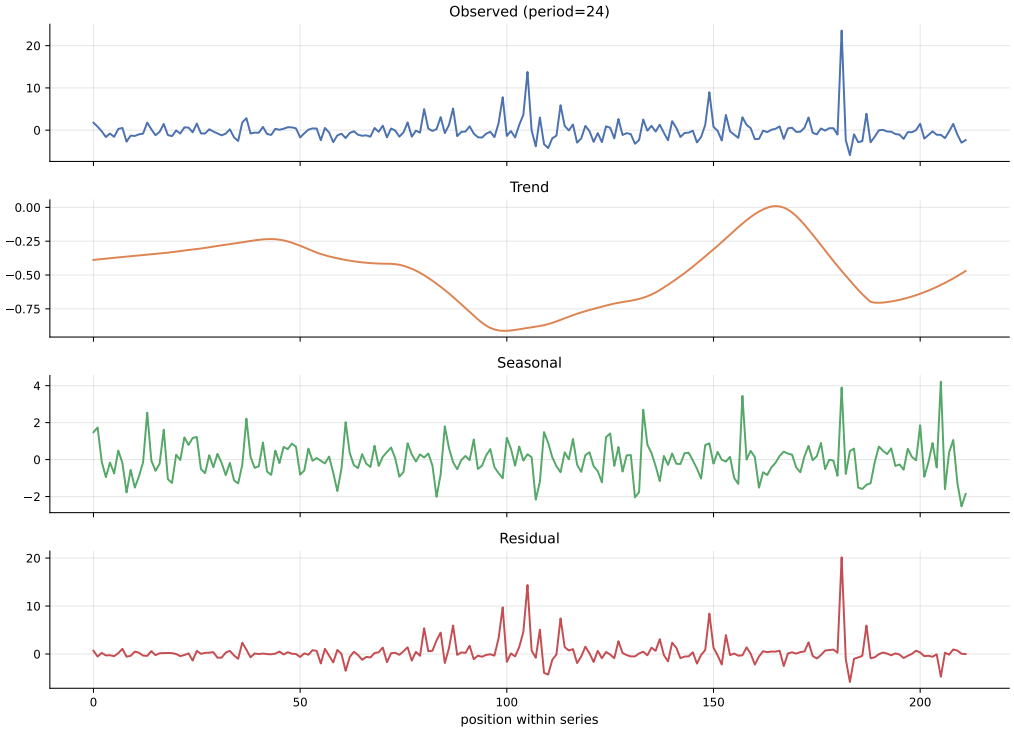

Figure 13: STL decomposition of the longest-history anchor with period 24. The trend component captures slow drift; the seasonal amplitude is small relative to residual standard deviation.

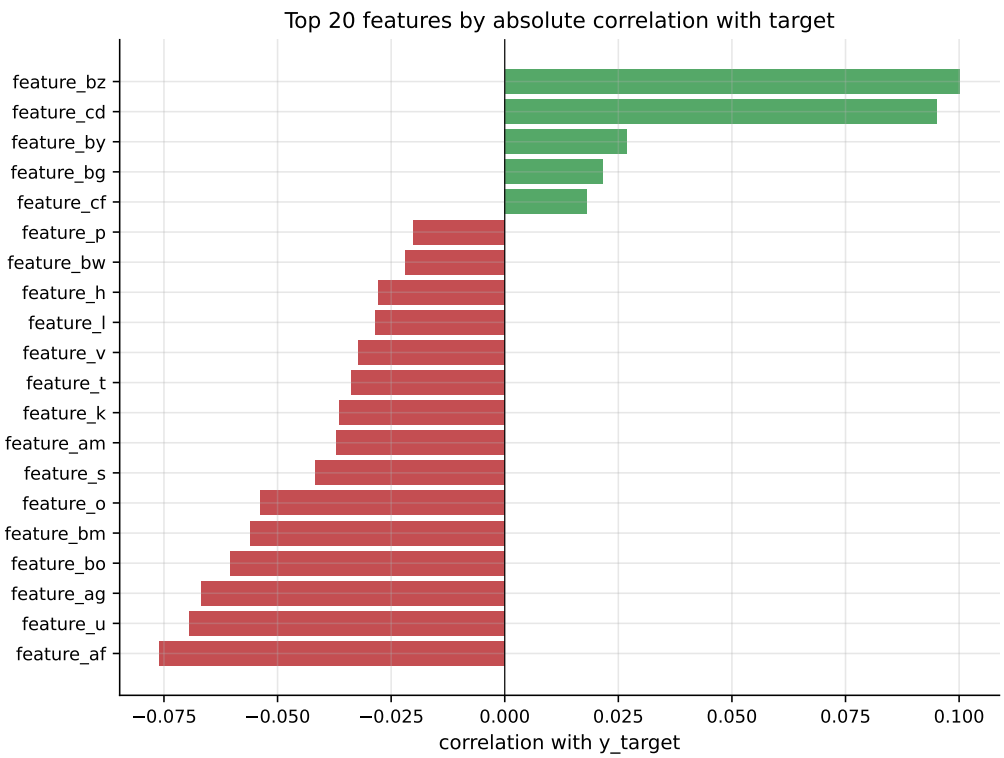

Figure 14: Top twenty features by absolute correlation with y_{target}, computed on a 200,000-row stratified sample. Green bars are positive correlations, red bars are negative.

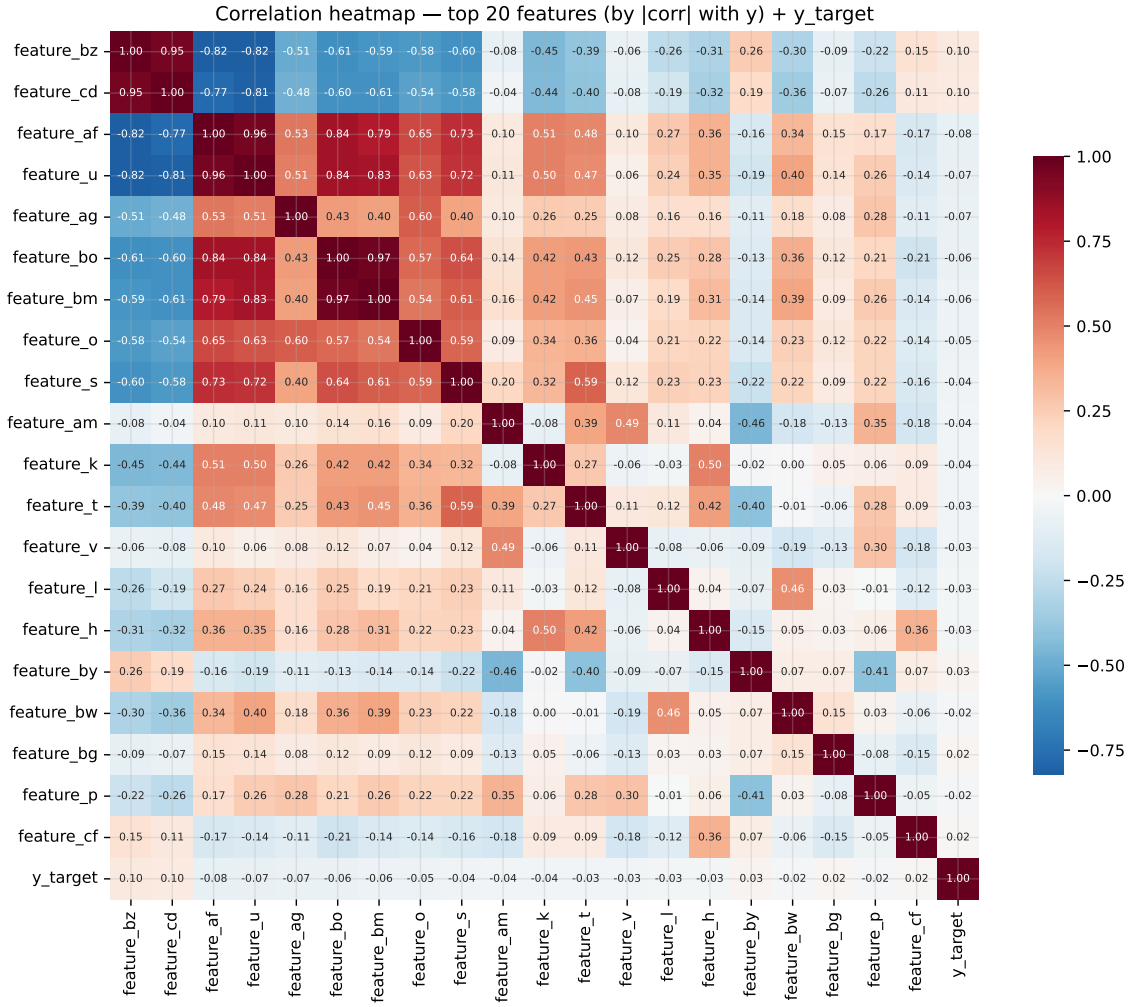

Figure 15: Correlation heatmap of the top twenty features (by |corr| with y_{target}) plus the target itself. Strong block structure indicates feature–feature redundancy.

Figure 16: Lagged cross-correlation between random sub-code pairs sharing the same code and horizon. A peak away from lag 0 means one sub-code leads or lags the other.

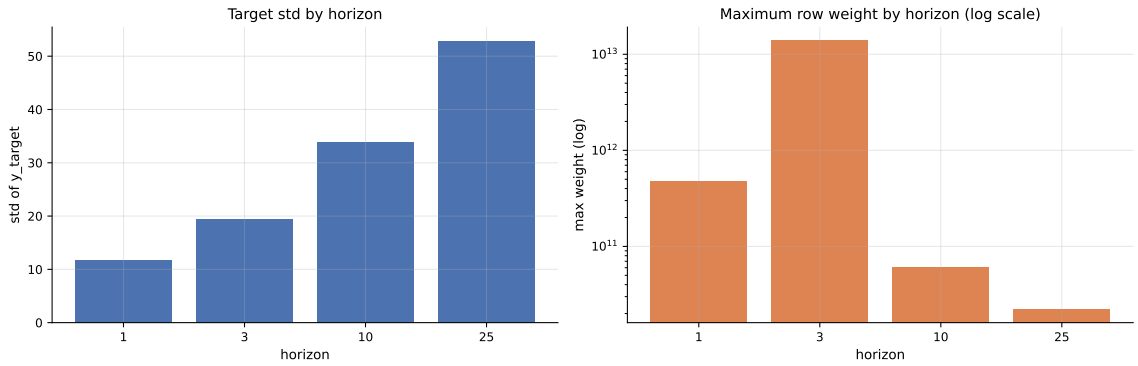

Figure 17: Per-horizon target standard deviation (left) and maximum row weight on a logarithmic scale (right). Target scale grows by a factor of approximately 4.5 from H1 to H25. Maximum weight is highest at H3.

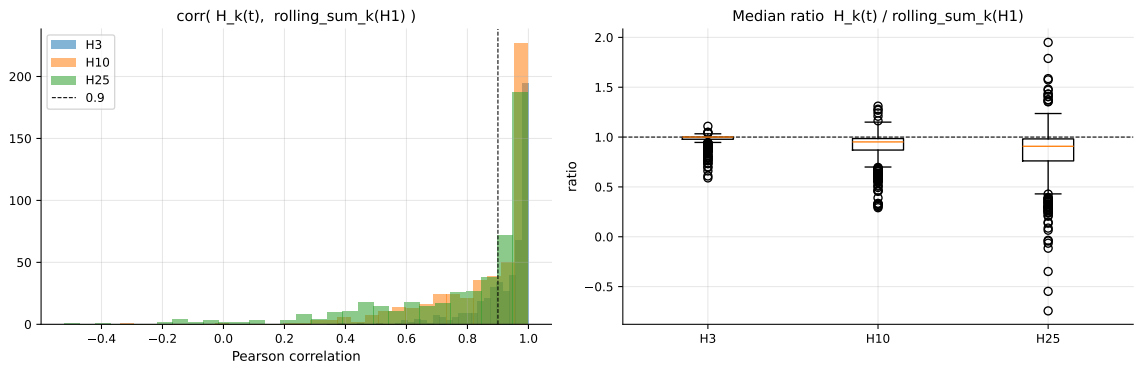

Figure 18: Test of the cumulation hypothesis on 500 stratified triples. Left: distribution of Pearson correlations between $y_t^{H_k}$ and the rolling sum of y^{H_1} over k steps, for $k \in \{3, 10, 25\}$. Right: distribution of the per-triple median ratio $y_t^{H_k} / \sum y^{H_1}$. Higher and tighter is better.

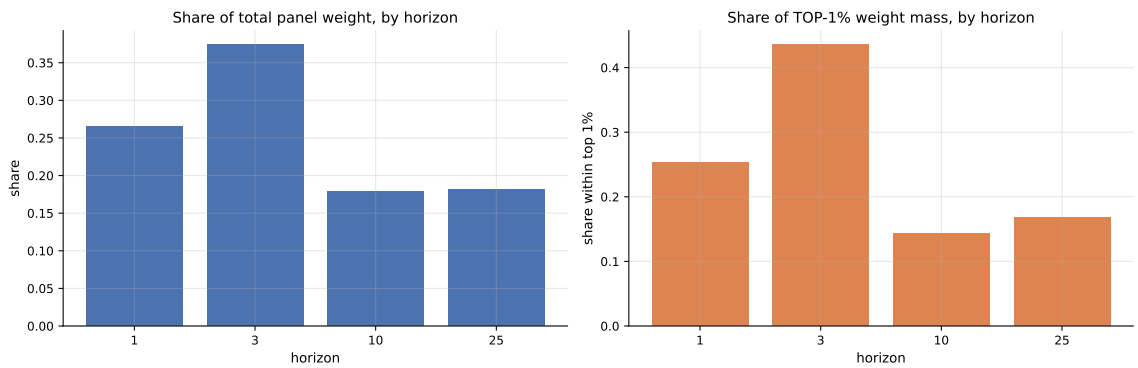

Figure 19: Per-horizon weight breakdown. Left: each horizon's share of total panel weight. Right: share of the top 1% of weight mass by horizon. H3 dominates both views.

Table 5: From EDA findings to modelling decisions.

EDA finding	**Modelling decision**
Test is a strict temporal suffix	Validation must be chronological. No random CV folds.
Top 1% of rows hold $\sim 64\%$ of weight	Sample-weight every loss; never report unweighted RMSE as headline.
Heavy target tails (± 2200) with IQR ± 0.05	Robust losses (Huber, quantile) for any model on raw y_{target}.
70% raw stationary, 99% after differencing	ARIMA with $d = 1$ default; differenced features for ML.
Persistent low-order ACF/PACF; weak STL seasonality	Lag features 1, 2, 3, 5, 10; no seasonal indicators.
Cross-series correlation within codes	Global pooled models over per-series fits.
Multi-horizon coupling and cumulation	Fit-H1-and-aggregate as a candidate model class. Per-horizon training as a baseline.
Per-horizon target scaling differs by $4.5\times$	One model per horizon, or horizon-aware target standardisation.
H3 dominates top-1% weight mass	Report H3 weighted skill as a separate headline.
Max $\lvert\text{corr}\rvert$ across features < 0.10	Gradient-boosted ensembles preferred over linear models on raw features.
Strong feature–feature correlation in top features	Read importance plots with collinearity in mind.